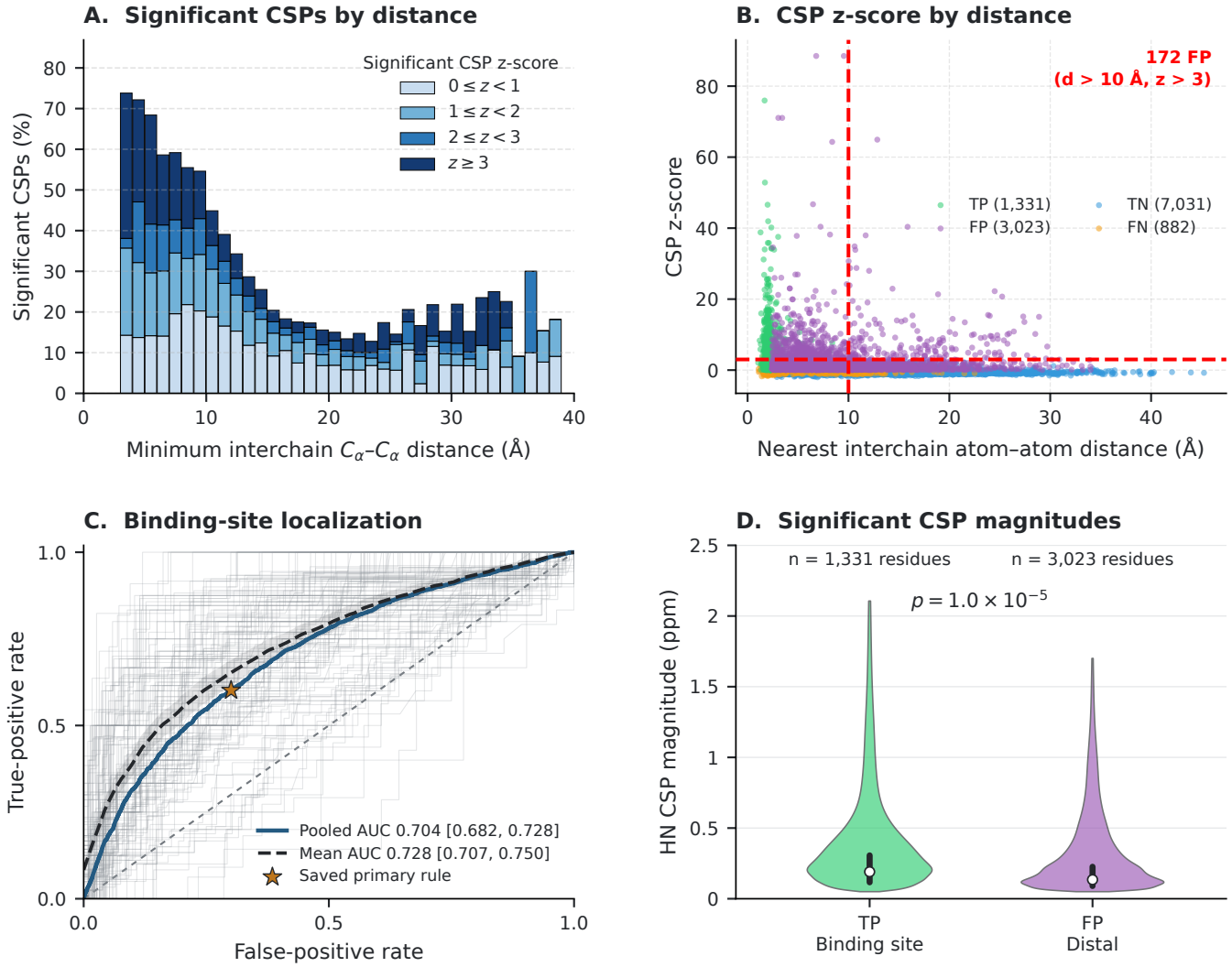


SI Fig. S17. CSPs and structural binding-site localization. All panels use 135 pH- and temperature-matched apo/holo targets. The per-target cutoff is max(cleaned mean HN CSP, 0.05 ppm). **A.** Percentage of residues with a significant HN CSP among those with a recorded CSP in each 1 Å bin of minimum interchain C_α–C_α distance. Stacks partition the significant fraction by CSP z-score. **B.** Per-residue CSP z-score versus nearest interchain atom–atom distance. For residue i, $z_i = (\mathrm{CSP}_i - \mu_{\mathrm{clean}})/\sigma_{\mathrm{clean}}$, where μ_{clean} and σ_{clean} are the mean and population standard deviation of the target's HN CSP list after iterative removal of high outliers. Colors indicate true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) relative to the structural binding-site definition. Dashed lines mark $z = 3$ and $d = 10$ Å. The 172 annotated FPs exceed both thresholds. **C.** Receiver operating characteristic (ROC) curves across HN CSP thresholds for 12,267 eligible residues. Gray curves show individual targets, blue shows the pooled ROC using CSP divided by the target-specific cutoff, and the black dashed curve shows the equal-target mean ROC. The pooled area under the curve (AUC) is 0.704 (95% CI 0.682–0.728), and the mean per-target AUC is 0.728 (95% CI 0.707–0.750). The band shows the pointwise 95% interval for the mean ROC. Intervals use 2,000 whole-target bootstrap replicates. The gray diagonal indicates chance, and the star marks the pooled operating point from the saved primary significance calls. **D.** HN CSP magnitudes for 1,331 significant binding-site residues (TP, green) and 3,023 significant distal residues (FP, purple) under the saved primary rule, pooled across targets. Scott-bandwidth kernel densities are estimated and normalized to equal area in $\log_{10}$ CSP space and bounded by the observed ranges. White dots mark medians and black bars mark interquartile ranges. The y-axis is linear (ppm). The symmetric Kullback–Leibler divergence was 0.0941 nats (20 shared log-spaced bins, 0.5 pseudocount per bin), with $p = 1.0 \times 10^{-5}$ from 99,999 within-target label permutations.